

STACK

Q4) Implement Two Stacks In Array

DS	Session	Stack	Question Information	Level 1	Challenge 44
Problem Description: Dr. Malar was booking a tour package of IRCTC from Chennai to Delhi for his family. Two of the relatives was interested in joining to this tour. these two persons are studying engineering in computing technology. only one tickets are remaining in the IRCTC portal. So, Dr. Malar decided to book one ticket for out of those persons also along with his family members. she wants to identify the one person out of these persons. he decided to conduct a technical task to identify the right person to travel. the task was that, implement two stack operations in an array Can you help them to complete the task? Constraints $0 < n < 5$ only five elements has to be practiced for this operation first element pushed into stack1 second element pushed into stack2, likewise elements pushed into alternative stacks vice versa. Function Description • Create a data structure <i>twoStacks</i> that represents two stacks. • Implementation of <i>twoStacks</i> should use only one array, i.e., both stacks should use the same array for storing elements. • Following functions must be supported by <i>twoStacks</i>. - push1(int x) -> pushes x to first stack - push2(int x) -> pushes x to second stack - pop1() -> pops an element from first stack and return the popped element					

CODE:

```
C main.c > twoStacks > top1
1  #include <stdio.h>
2
3  struct twoStacks {
4      int arr[10];
5      int top1;
6      int top2;
7  };
8
9  void push1(int x, struct twoStacks* ts) {
10     if (ts->top1 < ts->top2 - 1) {
11         ts->arr[++ts->top1] = x;
12     }
13 }
14
15 void push2(int x, struct twoStacks* ts) {
16     if (ts->top1 < ts->top2 - 1) {
17         ts->arr[--ts->top2] = x;
18     }
19 }
20
21 int pop1(struct twoStacks* ts) {
22     if (ts->top1 >= 0) {
23         return ts->arr[ts->top1--];
24     }
25     return -1;
26 }
27
28 int pop2(struct twoStacks* ts) {
29     if (ts->top2 < 10) {
30         return ts->arr[ts->top2++];
31     }
32     return -1;
33 }
34
35 int main(void) {
36     struct twoStacks ts;
37     ts.top1 = -1;
38     ts.top2 = 10;
39
40     int val;
41     for (int i = 0; i < 5; i++) {
42         if (scanf(format: "%d", &val) == 1) {
43             if (i % 2 == 0) {
44                 push1(x: val, ts: &ts);
45             } else {
46                 push2(x: val, ts: &ts);
47             }
48         }
49     }
50
51     printf(format: "Popped element from stack1 is:%d\n", pop1);
52     printf(format: "Popped element from stack2 is:%d\n", pop2);
53
54     return 0;
55 }
```

OUTPUT:

```
● [suraj@suraj-83gs test]$ ./main
1
2
3
4
5
Popped element from stack1 is:5
Popped element from stack2 is:4
○ [suraj@suraj-83gs test]$
```